

Soft-decision detection of weak tonals for passive sonar using track-before-detect method

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ABSTRACT

Tonal detection represents a significant task to detect and classify target sources for passive sonar in marine environments. However, in low signal-to-noise ratio (SNR) situations, tonals are difficult to detect and are readily immersed in the time-varying marine noise. The birth and death of tonals are also difficult to resolve due to bad detection performances. This paper utilizes the particle filter track-before-detect (PF-TBD) method to detect weak tonals with soft-decision results. The statistical properties of the tonal and noise are derived to build dynamic model, measurement model and likelihood function for tonal detection with TBD. The maximum likelihood estimation (MLE) method is used to estimate the noise variance which varies with time. A soft-decision detection result between 0 and 1 is estimated over each frequency bin to obtain an intuitive time–frequency spectrogram. Both the computer simulations and real sea trial data analyses (detect tonals from a deterministic source and a surface vessel) results validate the feasibility of the proposed method.

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1. Introduction

Acoustic radiation from vessels consists of the continuous spectrum and narrowband discrete components [1–4]. The narrowband discrete components, known as tonals, have a stronger stability and higher power than other radiated components [5,6]. Acoustic radiation from underwater targets are always rich in tonals. The tonals are one of the reliable and important features of the underwater targets. The target can be detected through the tonal detection by passive sonar. Therefore, detecting tonals embedded in a wide noise background is a significant task for passive sonar [7,8], especially to detect and classify targets over long distances. However, challenges still exist for tonal detection. First, improvements in noise reduction technique have made targets of interest quieter [9], which makes tonals too weak to be detected, especially over long distances. Second, the passive sonar receives time-varying noise from the marine environment, which can readily drown out the tonals [10]. Third, the number of tonals and their birth and death are unknown and varying for each target during the observation period.

The low-frequency analysis and recording (LOFAR) technique is used to detect tonals in current passive sonar systems [7]. The received signal is processed using the discrete Fourier transform (DFT), whose results, known as the amplitude spectrum in the time–frequency domain, are sent to a tonal detector [11]. A constant false alarm rate detector is used in the conventional approach [7,12–19], which gives detection results based on hypothesis testing. This results in hard-decision detection, which is 1 if the test statistical value exceeds the given threshold and is 0 otherwise. However, when the SNR of a target is low, the conventional detection performance degrades. To achieve a high detection probability, the threshold is set to low, which gives more false alarms. In this case, random births and deaths of targets are difficult to distinguish due to the high false alarm probability.

The track-before-detect (TBD) method was proposed to achieve better detection performances for weak targets [20]. This method takes non-thresholded measurements as the input data and avoids separating the detection and tracking processes [21,22]. In recent years, the particle filter (PF)-TBD method is developed and widely used in the fields of infrared and radar weak target detection and tracking [23–34]. This method is based on Bayesian theory and is suitable for non-Gaussian and non-linear situations. The PF-TBD approach has been applied in sonar to detect and track weak underwater targets [35–43], and for bearing-only tracking

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[42,43]. Yi focused on underwater tracking of unknown and time-varying targets using a passive array system with bearing measurements [42]. An accurate likelihood function was established using parameter estimation of different probability density function (PDF) models. These measurement models were given by three different classical beamformers. The trajectory initiation and termination strategy were judged from the quality of the particle partition for the birth and death of each target. Northward focused on the bearing-only localization performance for two complex data sets [43]. The likelihood function of the PF was designed from the accumulation of the beamformer output on the particle state trajectory. The birth and death of targets were assigned by the operator in this research. In [43], Northward also mentioned the limitation of his proposed approach of PF-TBD for passive sonar bearing-only estimation: the likelihood function yielded good results for the data sets examined but alternative functions could be conceived. That means the likelihood function should be built by considering the properties of the tracking object in different situations. Thus, the previous PF-TBD in these two papers cannot be utilized directly for passive sonar tonal detection. The dynamic model, measurement model and likelihood function should be built thoroughly by considering the tonal properties.

We propose using the PF-TBD method for weak tonal detection. The statistical properties of the tonal and noise are derived to build dynamic model, measurement model and likelihood function for tonal detection with TBD. The likelihood function is built from the amplitude spectrum distributions of the noise and tonal after DFT. The noise variance in the likelihood function is estimated from the maximum likelihood estimates (MLE) in real-time to satisfy the time-varying noise. Distinct from the hard-decision, the detection results are given between 0 and 1 based on the particle existence state variable. Thus, the birth and death of the tonals can be presented by the soft-decision detection results over each frequency bin during the observation period. The soft-decision detection results are then given as an intuitive time-frequency spectrogram. To evaluate the detection performance of the proposed method, the receiving operating characteristic (ROC) curves of both the conventional Neyman-Pearson (NP) criterion and PF-TBD methods are plotted based on simulations. The results of processing real data from sea trials from two voyages (one for detecting tonals from a deterministic source and the other for detecting tonals from a surface vessel) also validate the feasibility of the proposed methodology.

This paper is organized as follows. In Section 2, we propose the tonal detection models of the TBD. The soft-decision detector of a tonal based on the PF-TBD method is proposed in Section 3. The simulation results are shown in Section 4, which summarizes the detection performance. In Section 5, sea trial results are used to validate the performance of the proposed method. Finally, we conclude in Section 6.

2. TBD models for tonal detection

To perform tonal detection with the TBD method, the statistical properties of the received signal should be derived to establish the dynamic model, measurement model and likelihood function. The received signal is always processed by the DFT to obtain the amplitude spectrum in the time-frequency domain. The DFT results are the measurements for the TBD method for tonal detection. In this section, the dynamic model, measurement model and likelihood function of the TBD for tonal detection are built taking into consideration the statistical properties of tonal and noise.

2.1. Dynamic model

The tonal frequency and amplitude are assumed to vary slowly in time. Denoting the tonal state as $X_k = [f_k, A_k]^T$, f_k and A_k are the frequency and amplitude of tonal at time k , respectively. The dynamic model is assumed to be:

$$X_k = X_{k-1} + Q_k \quad (1)$$

where $Q_k = [q_f, q_A]^T$, $k (k = 1, 2, \dots, K)$ is the time index and q_f and q_A are the random dynamic noise for the frequency and amplitude, respectively. To describe the tonal detection problem in the TBD, we define a tonal-existence indicator variable e . If the tonal is present at time k , there is $e_k = 1$; otherwise, $e_k = 0$. The tonal-existence indicator variable e_k is modeled as a two-state Markov chain with parameters:

$$P_{\text{birth}} = P(e_k = 1 | e_{k-1} = 0) \quad (2)$$

$$P_{\text{death}} = P(e_k = 0 | e_{k-1} = 1) \quad (3)$$

2.2. Measurement model and likelihood function

The input measurements of the TBD are non-thresholded; thus, at time k , the measurement can be described as:

$$Z_k = h(e_k X_k, g_k) \quad (4)$$

In (4), the function h shows the relationship at the time k between the measurement Z_k and the state X_k with the noise g_k . If the tonal is absent at the time k , $e_k = 0$ and the measurement model is $h(0, g_k)$; otherwise, $e_k = 1$ and the measurement model is $h(X_k, g_k)$.

In the optical and infrared fields, the function h is always denoted as a point spread function (PSF) [24,27,34]. However, to consider tonal detection, h is decided by the DFT process. Thus the measurement (4) can be rewritten as:

$$Z_k = h(e_k s_k + g_k) \quad (5)$$

where s_k is a single tonal with a frequency f_k and amplitude A_k in tonal state X_k :

$$s_k(n) = A_k \cos(2\pi f_k n) \quad n = 1, 2, \dots, N \quad (6)$$

and g_k is the wideband Gaussian noise with mean 0 and variance σ_g^2 .

The N -point DFT of $s_k(n)$ is:

$$\begin{aligned} S_k(f) &= \frac{1}{N} \sum_{n=0}^{N-1} s_k(n) e^{-j2\pi f n} \\ &= \frac{A_k}{2N} \cdot (e^{j\pi(f_k - f)(N-1)} \cdot \frac{\sin(N\pi(f_k - f))}{\sin(\pi(f_k - f))} + e^{-j\pi(f_k + f)(N-1)} \\ &\quad \cdot \frac{\sin(N\pi(f_k + f))}{\sin(\pi(f_k + f))}) \end{aligned} \quad (7)$$

To simplify the analysis, we only consider positive frequency values ($f > 0$). That is:

$$S_k(f) = \frac{A_k}{2N} e^{j\pi(f_k - f)(N-1)} \cdot \frac{\sin(N\pi(f_k - f))}{\sin(\pi(f_k - f))} \quad (8)$$

The N -point DFT of noise $g_k(n)$ is:

$$G_k(f) = \frac{1}{N} \sum_{n=0}^{N-1} g_k(n) e^{-j2\pi f n} = G_{R,k}(f) + jG_{I,k}(f) \quad (9)$$

where $G_{R,k}(f)$ and $G_{I,k}(f)$ are the real and imaginary parts of $G_k(f)$, respectively. As $G_k(f)$ is a linear combination of $g_k(n)$, both $G_{R,k}(f)$

and $G_{l,k}(f)$ follow Gaussian distributions. The means and variances of $G_{R,k}(f)$ and $G_{l,k}(f)$ are [7]:

$$E(G_{R,k}(f)) = E(G_{l,k}(f)) = 0 \quad (10)$$

$$D(G_{R,k}(f)) = D(G_{l,k}(f)) = \frac{1}{N^2} \sum_{n=0}^{N-1} \sigma_g^2 \cos^2(2\pi fn) = \frac{\sigma_g^2}{2N} \quad (11)$$

The $G_{R,k}(f)$ and $G_{l,k}(f)$ are independent since:

$$E(G_{R,k}(f)G_{l,k}(f)) = \frac{\sigma_g^2}{N^2} \sum_{n=0}^{N-1} \cos(2\pi fn) \sin(2\pi fn) = 0 \quad (12)$$

Denoting $y_k(n) = e_k s_k(n) + g_k(n)$. $Y_k(f)$ is the N -point DFT result of $y_k(n)$. Then, the distribution of $Y_k(f)$ can be denoted as $Y_k(f) \sim CN(0, 2\sigma_G^2)$ when $e_k = 0$ and $Y_k(f) \sim CN(S_k(f), 2\sigma_G^2)$ when $e_k = 1$, where $\sigma_G^2 = \frac{\sigma_g^2}{2N}$ and CN denotes the complex Gaussian distribution.

In the TBD method for tonal detection, the measurement is the amplitude spectrum at the time k at frequency f :

$$Z_k(f) = |Y_k(f)| \quad (13)$$

With the distribution of $Y_k(f)$, we know that $Z_k(f)$ follows a Rayleigh distribution when $e_k = 0$:

$$p(Z_k(f)|e_k = 0) = \frac{Z_k(f)}{\sigma_G^2} \exp\left(-\frac{Z_k(f)^2}{2\sigma_G^2}\right) \quad (14)$$

and a Rician distribution when $e_k = 1$:

$$p(Z_k(f)|X_k, e_k = 1) = \frac{Z_k(f)}{\sigma_G^2} \exp\left(-\frac{|S_k(f)|^2 + Z_k(f)^2}{2\sigma_G^2}\right) I_0\left(\frac{|S_k(f)|Z_k(f)}{\sigma_G^2}\right) \quad (15)$$

where $I_0(\cdot)$ is the modified Bessel function of the first kind of order 0:

$$I_0(x) = \sum_{a=0}^{\infty} \frac{(x/2)^{2a}}{a! \Gamma(a+1)} \quad (16)$$

Then, we can get the likelihood ratio function with (14) and (15):

$$\begin{aligned} l(Z_k(f)|X_k, e_k) &= \frac{p(Z_k(f)|X_k, e_k = 1)}{p(Z_k(f)|e_k = 0)} \\ &= \exp\left(-\frac{|S_k(f)|^2}{2\sigma_G^2}\right) I_0\left(\frac{|S_k(f)|Z_k(f)}{\sigma_G^2}\right) \end{aligned} \quad (17)$$

Assuming the processing band has M frequency bins, m ($m = 1, 2, \dots, M$) is the frequency index of f . Namely, $Z_k(f)$ is equal to $Z_k^{(m)}$. The set of measurements at time k is $\mathbf{Z}_k = (Z_k^{(1)}, Z_k^{(2)}, \dots, Z_k^{(M)})^T$. Then, we have:

$$\begin{aligned} l(\mathbf{Z}_k|X_k, e_k) &= \prod_{b=1}^M \frac{p(Z_k^{(b)}|X_k, e_k = 1)}{p(Z_k^{(b)}|e_k = 0)} \\ &= \prod_{b=1}^M \exp\left(-\frac{(\hat{Z}_k^{(b)})^2}{2\sigma_G^2}\right) I_0\left(\frac{Z_k^{(b)} \hat{Z}_k^{(b)}}{\sigma_G^2}\right) \end{aligned} \quad (18)$$

In (18), the noise variance σ_G^2 should be estimated in real-time as it is unknown and time-varying in real sea detection situations. The MLE theory can be studied to estimate the value of σ_G^2 . The estimated tonal is considered in the m th frequency bin at time k , while the others are the noise bins. The value of σ_G^2 can be estimated in real-time from the noise bins at time k . The PDF of $\mathbf{Z}_k$ when $e_k = 0$ is:

$$p(\mathbf{Z}_k|e_k = 0) = \prod_{b=1}^{m-1} p(Z_k^{(b)}|e_k = 0) \prod_{b=m+1}^M p(Z_k^{(b)}|e_k = 0) \quad (19)$$

The partial derivative of the logarithm of $p(\mathbf{Z}_k|e_k = 0)$ is:

$$\begin{aligned} \frac{\partial \ln(p(\mathbf{Z}_k|e_k = 0))}{\partial \sigma_G^2} &= -\frac{M-1}{\sigma_G^2} + \frac{1}{2\sigma_G^4} \left(\sum_{b=1}^{m-1} (Z_k^{(b)})^2 \right. \\ &\quad \left. + \sum_{b=m+1}^M (Z_k^{(b)})^2 \right) \end{aligned} \quad (20)$$

Equating (20) to zero, we find the MLE estimation of σ_G^2 is:

$$\hat{\sigma}_G^2 = \frac{1}{2(M-1)} \left(\sum_{b=1}^{m-1} (Z_k^{(b)})^2 + \sum_{b=m+1}^M (Z_k^{(b)})^2 \right) \quad (21)$$

For practical applications, multi-tonal situations should be considered. Thus, the noise data should be collected over a reference window with a reasonable length on both sides of the m th frequency bin. A protection window is also necessary to avoid obtaining the tonal data in neighboring bins. Then, the noise variance at time k can be estimated:

$$\hat{\sigma}_G^2 = \begin{cases} \frac{1}{2W_2} \left(\sum_{b=m-W_1+1}^{m+W_2} (Z_k^{(b)})^2 \right), & 0 < m \leq W_2 \\ \frac{1}{2(W_2-W_1+m-1)} \left(\sum_{b=1}^{m-W_1-1} (Z_k^{(b)})^2 + \sum_{b=m+W_1+1}^{m+W_2} (Z_k^{(b)})^2 \right), & W_2 < m \leq W_1 + W_2 \\ \frac{1}{4W_2} \left(\sum_{b=m-W_1-1}^{m-W_2} (Z_k^{(b)})^2 + \sum_{b=m+W_1+1}^{m+W_2} (Z_k^{(b)})^2 \right), & W_1 + W_2 < m \leq M - W_1 - W_2 \\ \frac{1}{2(W_2-W_1+M-m-1)} \left(\sum_{b=m-W_1-1}^{m-W_2} (Z_k^{(b)})^2 + \sum_{b=m+W_1+1}^M (Z_k^{(b)})^2 \right), & M - W_1 - W_2 < m \leq M - W_2 \\ \frac{1}{2W_2} \left(\sum_{b=m-W_1-1}^{m-W_2} (Z_k^{(b)})^2 \right), & M - W_2 < m \leq M \end{cases} \quad (22)$$

where W_1 is the length of the protection window and W_2 is the length of the reference window.

3. Soft-decision detector of tonal by PF-TBD

For the TBD approach, an approximate method is needed to accomplish the specific estimation of the target states. The particle filter method is a versatile and effective computational approach to solve non-linear non-Gaussian Bayesian filtering models and is a reasonable implementation choice for the Bayesian TBD, namely PF-TBD.

In the PF-TBD method, each particle has a tonal-existence indicator variable e_k to represent the existence state of the target. The particle indicator variable also evolves as a Markov chain with parameters (2) and (3). According to Bayesian theory, the target state at time k can be described as a posterior density [34,35,39,40]:

$$\begin{aligned} p(X_k, e_k = 1|\mathbf{Z}_k) &= \frac{p(\mathbf{Z}_k|X_k, e_k = 1)p(X_k, e_k = 1|\mathbf{Z}_{k-1})}{p(\mathbf{Z}_k|\mathbf{Z}_{k-1})} \\ &\propto l(\mathbf{Z}_k|X_k, e_k) p(X_k, e_k = 1|\mathbf{Z}_{k-1}) \end{aligned} \quad (23)$$

In (23), $p(X_k, e_k = 1|\mathbf{Z}_{k-1})$ is:

$$\begin{aligned} &= \int p(X_k, e_k = 1|\mathbf{Z}_{k-1}, e_{k-1} = 1) p(\mathbf{Z}_{k-1}, e_{k-1} = 1|\mathbf{Z}_{k-2}) d\mathbf{X}_{k-1} + p(X_k, e_k = 1, e_{k-1} = 0|\mathbf{Z}_{k-1}) \\ &= \int p(X_k|X_{k-1}, e_k = 1, e_{k-1} = 1) (1 - P_{\text{death}}) p(X_{k-1}, e_{k-1} = 1|\mathbf{Z}_{k-1}) d\mathbf{X}_{k-1} + p_0(X_k) P_{\text{birth}} \end{aligned} \quad (24)$$

In (24), $p_0(X_k)$ is the initial prior density. The posterior density at time k (23) is given by the likelihood ratio function (18), the dynamic model (1)-(3), and the posterior density at the time $k-1$.

3.1. PF-TBD implementation

To detect and track multiple tonals, a PF-TBD filter is set to keep estimating the states and obtain the detection results for each fre-

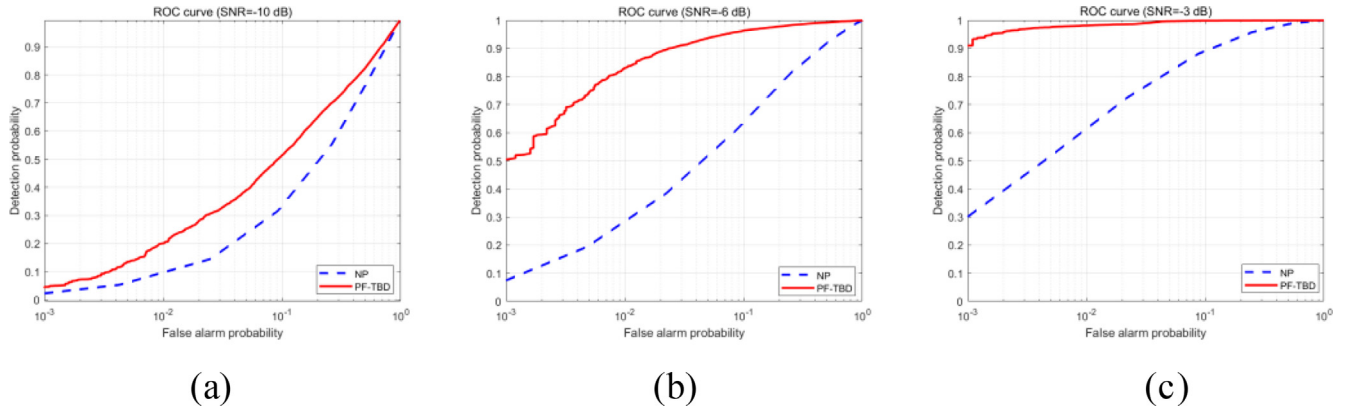


Fig. 1. ROC curves of the NP and PF-TBD detectors at SNRs of (a) -10 (b) -6 (c) -3 dB.

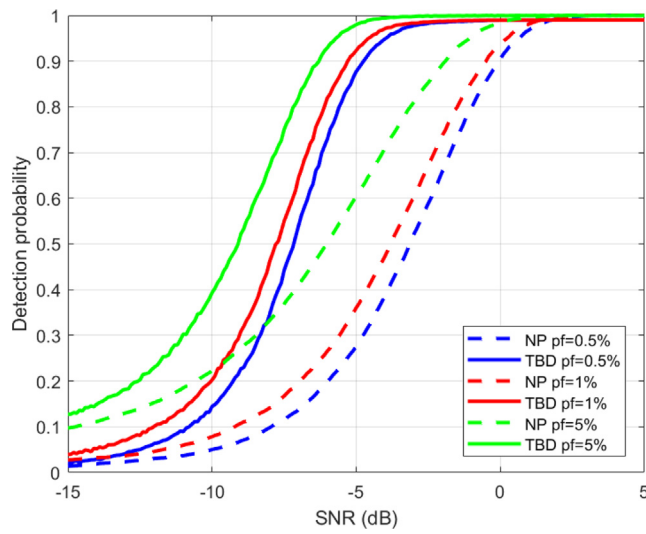


Fig. 2. Detection probabilities of the NP detector and PF-TBD at false alarm probabilities of 0.5%, 1%, and 5%.

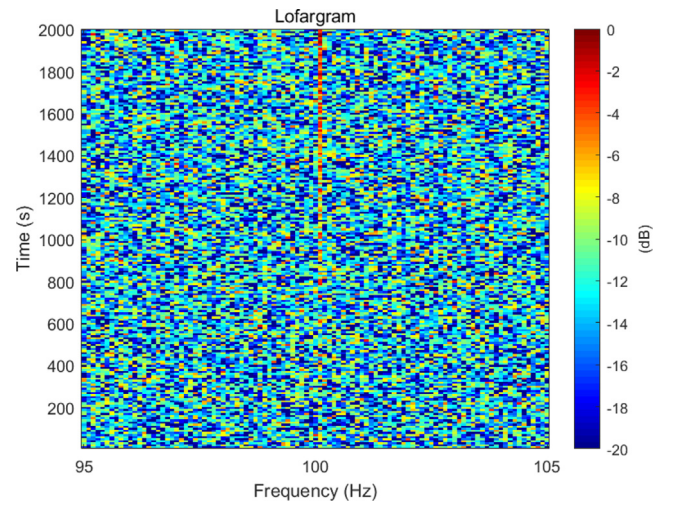
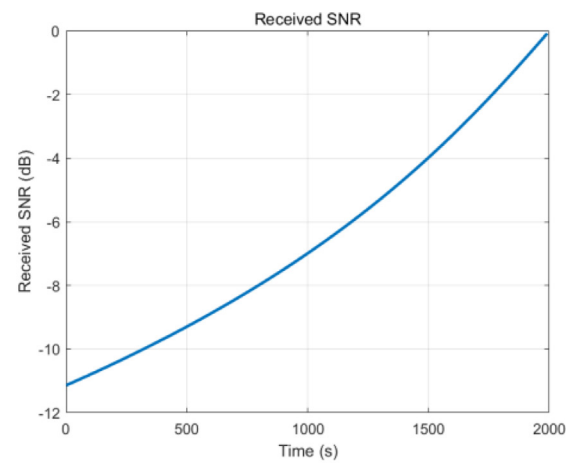
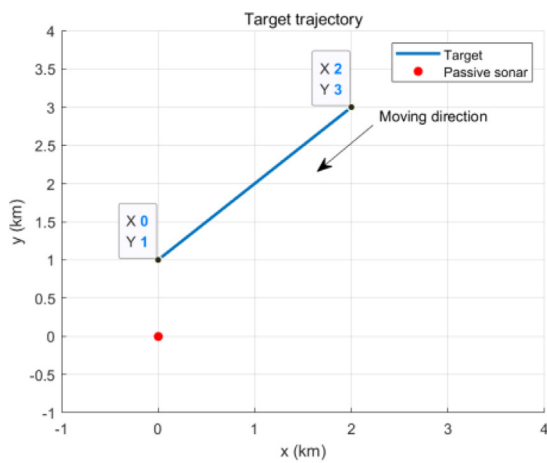


Fig. 4. Lofargram during the observation period.



(a)

(b)

Fig. 3. (a) Target trajectory and (b) SNR of the target.

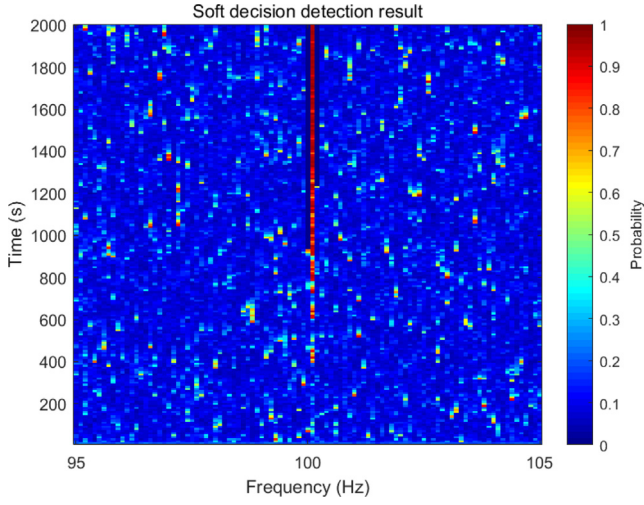


Fig. 5. Soft-decision detection results from the PF-TBD.

quency bin. If there are M bins, there are M filters, which work together to give the detection and estimation results to each tonal.

3.1.1. Initialization ($k = 0$)

Assume there are N_p particles; the i th particle's initial state and weight for the m th frequency bin are $(X_0^{(m)i}, e_0^{(m)i})$ and $w_0^{(m)i}$, respectively, where:

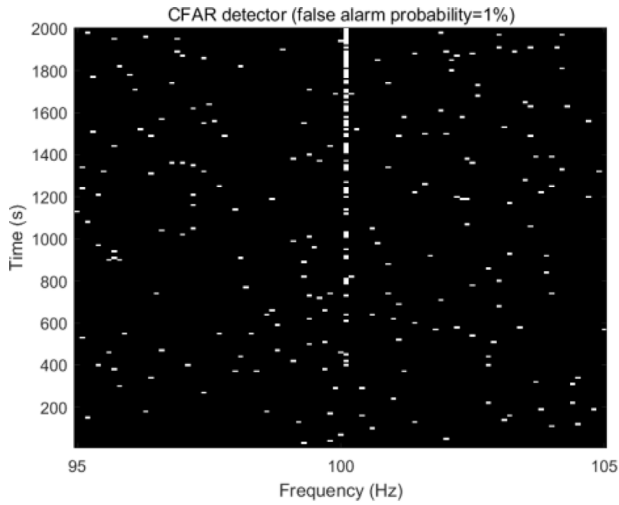
$$X_0^{(m)i} \sim p_0(X_0^{(m)}) \quad (25)$$

$$e_0^{(m)i} = \begin{cases} 0, & 1 - P_{\text{birth}} \\ 1, & P_{\text{birth}} \end{cases} \quad (26)$$

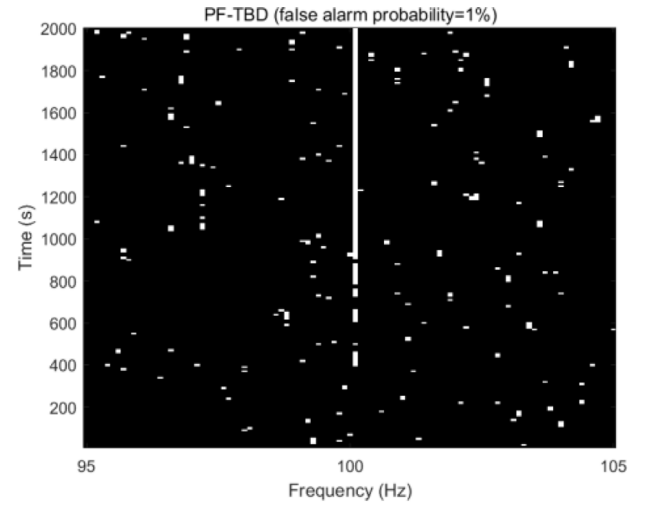
$$w_0^{(m)i} = \frac{1}{N_p} \quad (27)$$

3.1.2. State prediction

When $k > 0$, the state of each particle is updated. The prediction of the tonal-existence indicator variable $e_k^{(m)i}$ follows a Markov pro-

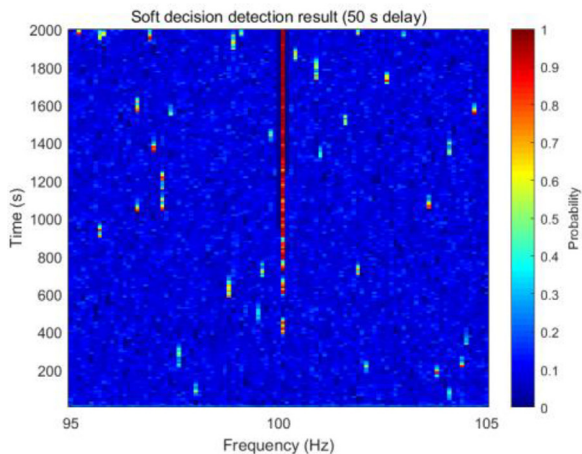


(a)

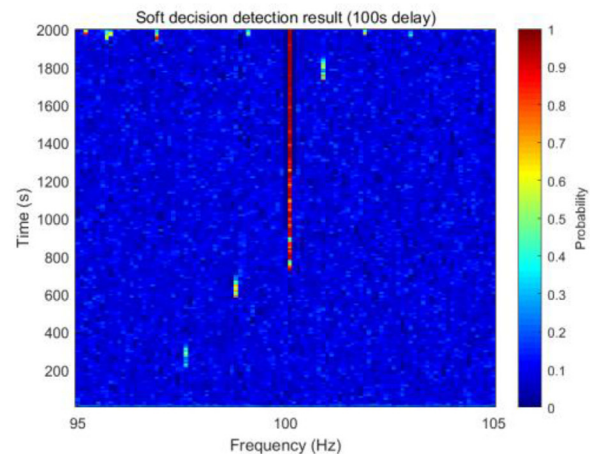


(b)

Fig. 6. Detection results of the (a) NP detector and (b) PF-TBD when the false alarm probability is 1%.



(a)



(b)

Fig. 7. PF-TBD soft-decision detection results with unreal-time processing with delay time (a) 50 and (b) 100 s.

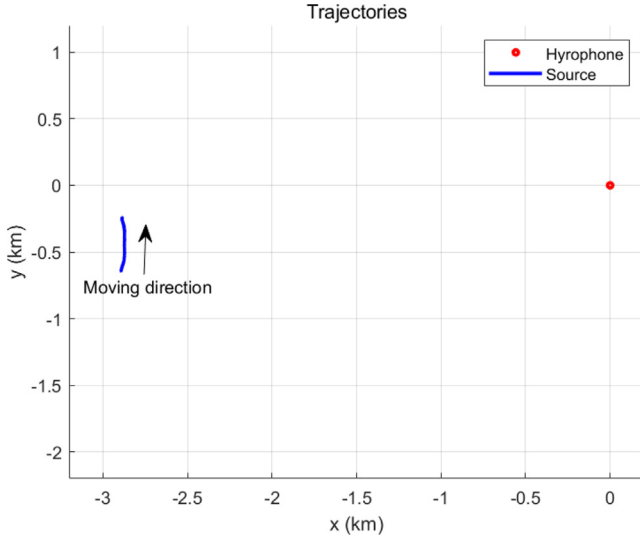


Fig. 8. GPS trajectories of the two boats.

cess with parameters given by (2) and (3). The prediction of $X_k^{(m)i}$ is divided into two parts. For particles of tonals newly born at time k :

$$X_k^{(m)i} \sim p(X_k | e_k = 1, e_{k-1} = 0, \mathbf{Z}_k) \quad (28)$$

For particles of previously existing tonals:

$$X_k^{(m)i} \sim p(X_k | X_{k-1}, e_k = 1, e_{k-1} = 1, \mathbf{Z}_k) \quad (29)$$

The $p(X_k | X_{k-1}, e_k = 1, e_{k-1} = 1, \mathbf{Z}_k)$ is obtained from the dynamic model (1), and $p(X_k | e_k = 1, e_{k-1} = 0, \mathbf{Z}_k)$ follows the initialization step 1.

3.1.3. Particle weight

The weight of each particle can be calculated from the likelihood ratio function (17):

$$\tilde{w}_k^{(m)i} = w_{k-1}^{(m)i} l(\mathbf{Z}_k | X_k^{(m)i}, e_k^{(m)i} = 1) = w_{k-1}^{(m)i} \frac{p(\mathbf{Z}_k | X_k^{(m)i}, e_k^{(m)i} = 1)}{p(\mathbf{Z}_k | e_k^{(m)i} = 0)} \quad (30)$$

After normalization, the weight of each particle is:

$$w_k^{(m)i} = \frac{\tilde{w}_k^{(m)i}}{\sum_{i=1}^{N_p} \tilde{w}_k^{(m)i}} \quad (31)$$

3.1.4. Resampling

The resampling step aims to restrain the degeneration of particles. The results of the prediction are a set of weighted particles in the state space, which represents the posterior. In the resampling step, we obtain a representation of the posterior with equal-weighted particles by deleting those with lighter weights and replicating heavier ones. Then, the number of particles remains the same and the weight of each particle becomes $\hat{w}_k^{(m)i} = \frac{1}{N_p}$. The state of the i th particle after resampling is $(\hat{X}_k^{(m)i}, \hat{e}_k^{(m)i})$.

3.1.5. Estimation

The particles with indicator variable $e_k^{(m)i} = 1$ can present posterior density of the state of tonal approximately:

$$p(X_k^{(m)}, e_k^{(m)i} = 1 | \mathbf{Z}_k) \approx \sum_{i=1}^{N_p} w_k^{(m)i} \delta(X_k^{(m)} - X_k^{(m)i}) e_k^{(m)i} \quad (32)$$

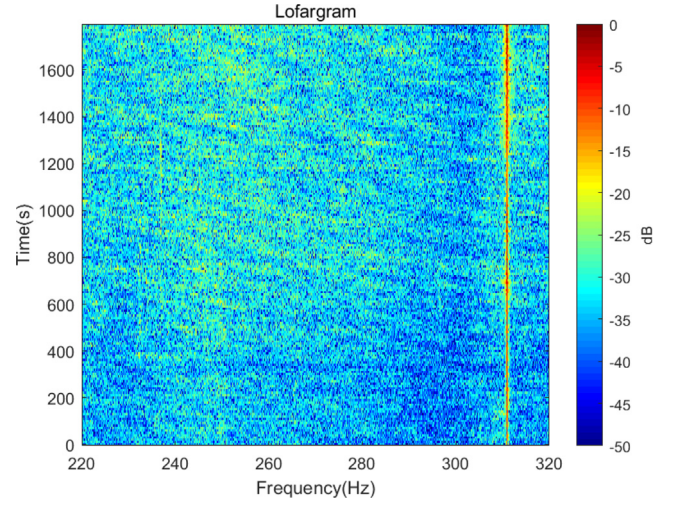


Fig. 9. Lofargram after the DFT during the observation period from the first voyage.

And the tonal's state is estimated from the particles with $\hat{e}_k^{(m)i} = 1$:

$$X_k^{(m)} = \frac{1}{\sum_{i=1}^{N_p} \hat{e}_k^{(m)i}} \sum_{i=1}^{N_p} \hat{X}_k^{(m)i} \hat{e}_k^{(m)i} \quad (33)$$

3.2. Soft-decision detection by PF-TBD

In a classic detection and tracking process, the detection requires a fixed or adaptive threshold to obtain measurements as potential tonals, which is called a hard-decision. Hard-decision is a binary hypotheses test and only a binary result is given: tonal is present or not. In a TBD process, all measurements are used without thresholding. The likelihood function is derived from the statistical properties of the noise and tonal. In the PF-TBD, the number of particles is fixed at each time, and the tonal-existence indicator variable $e_k^{(m)i}$ represents the existence state of the tonal. The ratio of the number of particles with $e_k^{(m)i} = 1$ to the total particle number N_p is between 0 and 1, which can be described as a soft-decision result at time k .

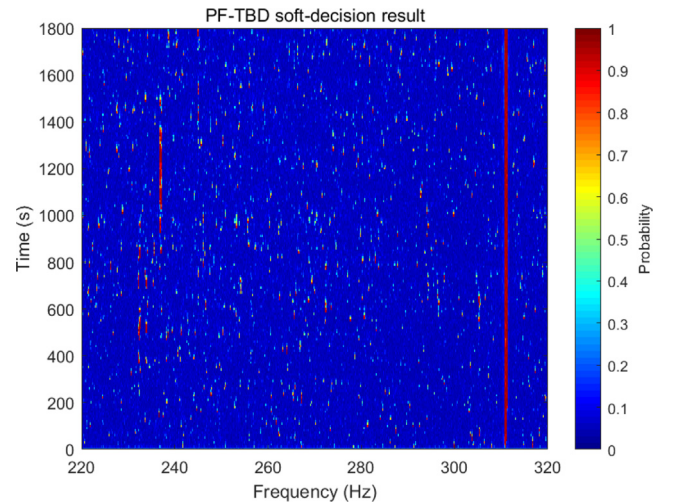


Fig. 10. Corresponding PF-TBD soft-decision detection results from Fig. 9.

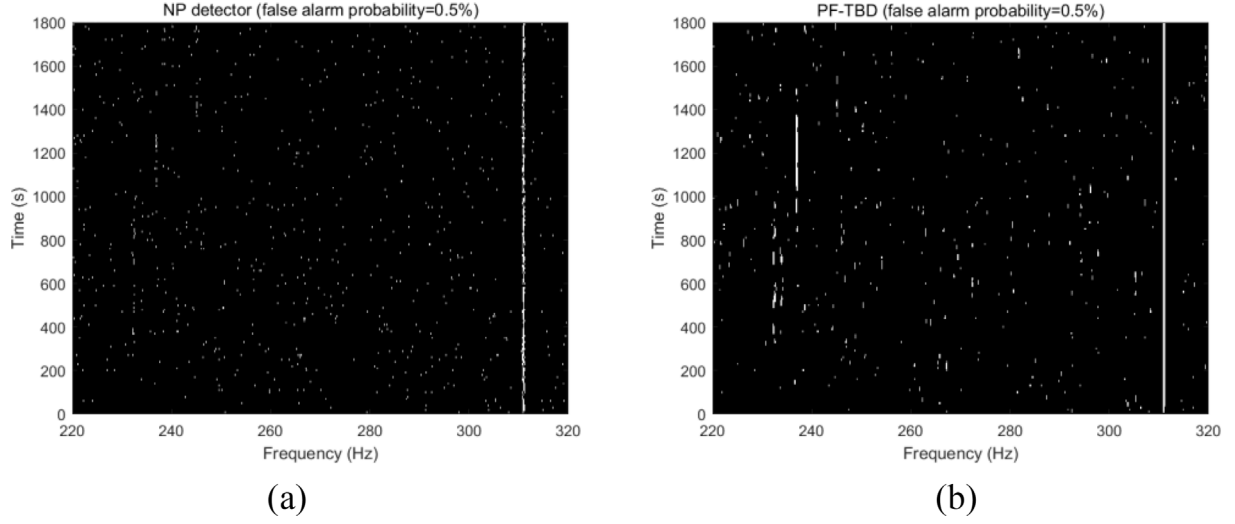


Fig. 11. Detection results of the (a) NP detector and (b) PF-TBD when the false alarm probability is 0.5%.

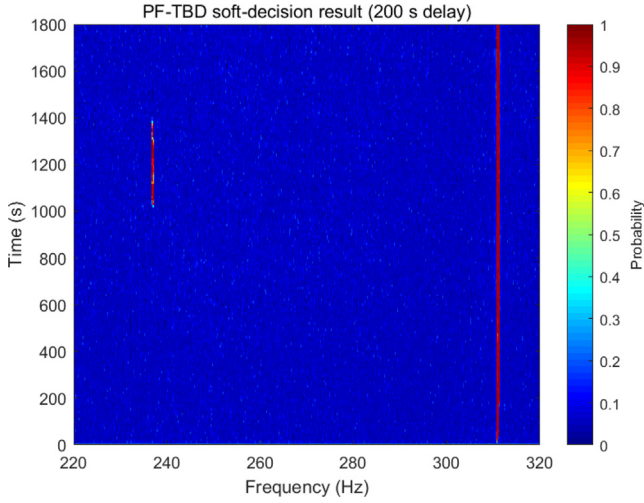
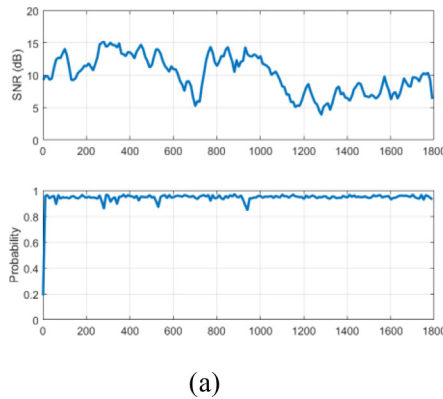


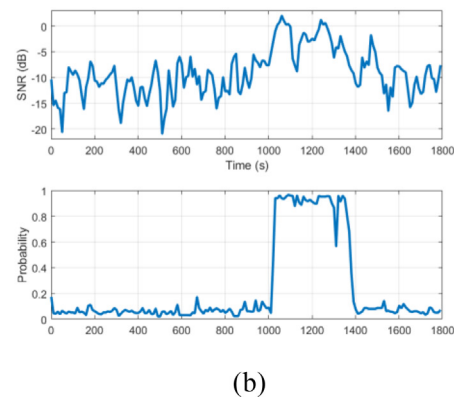
Fig. 12. PF-TBD soft-decision detection results with unreal-time processing with delay time 200 s.

We define the tonal-existence probability at the m th frequency bin $P_{e,k}^{(m)}$ as:

$$P_{e,k}^{(m)} = \frac{\sum_{i=1}^{N_p} \hat{e}_k^{(m)i}}{N_p} \quad (34)$$



(a)



(b)

Fig. 13. The SNR and PF-TBD soft-decision detection results at (a) 311 Hz and (b) 237 Hz.

At the time k , $P_{e,k}^{(m)}$ shows the probability of tonal-existence as a number between 0 and 1 instead of a direct hard-decision. For all M frequency bins and K times, the time–frequency spectrogram shows the soft-decision detection results in the frequency band of interest during the observation period. For each tonal track, a continuous soft-decision detection result is given at the observation time period. The birth and death are shown by the tonal-existence probability.

4. Simulations

This section presents the performances of the proposed PF-TBD method for tonal detection. First, the ROC curves of the detector based on NP criterion and the proposed method are given to compare their detection performances. Second, a moving target is simulated and detected using passive sonar. The detection results show that under the same false alarm probability, the proposed method has a higher and more steady detection performance.

In the simulations, the SNR is given by:

$$SNR = 10 \log_{10} \left(\frac{A^2}{2\sigma_s^2} \right) \quad (35)$$

The parameters of the PF-TBD in the simulations are as follows:

For the m th filter, the initial frequency is drawn from the uniform distribution as $f_0^{(m)} \sim U(f_0^{(m)} - \frac{\Delta f}{2}, f_0^{(m)} + \frac{\Delta f}{2})$, where $\Delta f = \frac{1}{T}$;

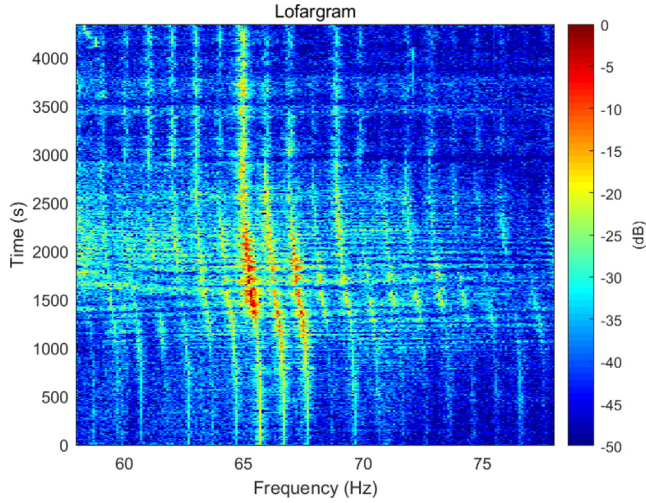


Fig. 14. Lofargram after the DFT during the observation period.

The value of the frequency state noise q_f follows a Gaussian distribution $q_f \sim N(0, \frac{\Delta f}{100})$;

The initial amplitude is drawn from the uniform distribution as $A_0^{(m)} \sim U(A_0^{(m)} - \Delta A, A_0^{(m)} + \Delta A)$, where $A_0^{(m)} = 2Z_0^{(m)}$ and $\Delta A = \frac{1}{M} \sum_{m=1}^M A_0^{(m)}$;

The value of amplitude state noise q_A follows a Gaussian distribution $q_A \sim N(0, \frac{1}{2} \Delta A)$;

The noise variance from the likelihood function σ_C^2 should be estimated from (22). In the simulations, we set the protection window 1 Hz and the reference window 5 Hz.

4.1. Detection performance

The detection results for the PF-TBD are soft-decision detection. Thus, the test statistical variable is the existence probability. The tonal is considered detected when the soft-decision detection results are above a tunable threshold [23,34,44]. Associated ROC curves can be generated by setting different thresholds and performing parametric experiments.

To compare the detection performances with the previous approaches, we consider detector with the NP criterion. The tonal signal is a sinusoidal signal on frequency 100 Hz. The inputs of the detectors are the DFT results. The DFT size is 10 s with a 15 kHz sample frequency. The noise follows a Gaussian distribution with a mean of 0 and variance of 1. The number of PF-TBD particles is 500, $P_{\text{birth}} = P_{\text{death}} = 0.05$. The ROC curves over 10000 experiments at SNRs of -10, -6 and -3 dB are shown in Fig. 1.



Fig. 15. Container vessel (photo from Ref. 45).

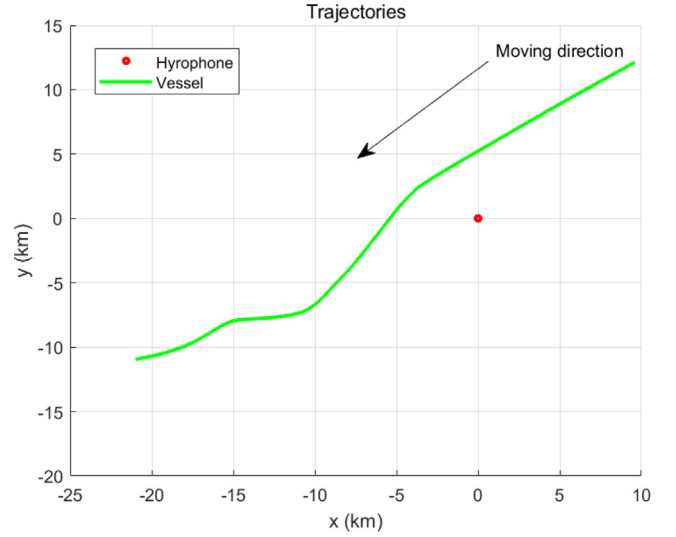


Fig. 16. Relative trajectory of the container vessel.

Fig. 1 shows that the detection probability of the PF-TBD is higher than the NP detector for any given false alarm probability.

Fig. 2 shows the detection probabilities of the two methods under different SNRs when the false alarm probability is fixed at 0.5%, 1%, and 5%. Compared to the NP detector, the PF-TBD has a 3 to 5 dB gain under the same false alarm probability. Thus, the detection performance of the PF-TBD is much better than the NP detector especially at low SNRs (below -3 dB). For example, when SNR is -6 dB, the detection probability of PF-TBD is nearly 50% higher than the NP if the false alarm probability is fixed at 1%.

4.2. Passive detection of a moving target

Consider a target moving in a straight line towards a passive sonar receiver at a constant speed [Fig. 3a]. The initial and final positions of the target are (2 km, 3 km) and (0 km, 1 km), respectively. The passive sonar is at the origin, the travel time is 2000 s. Assuming the target radiates a sinusoidal signal on frequency 100 Hz, and the tonal from the target is on 100 Hz. The noise is Gaussian with a mean of 0 and variance of 1.

The passive sonar equation is:

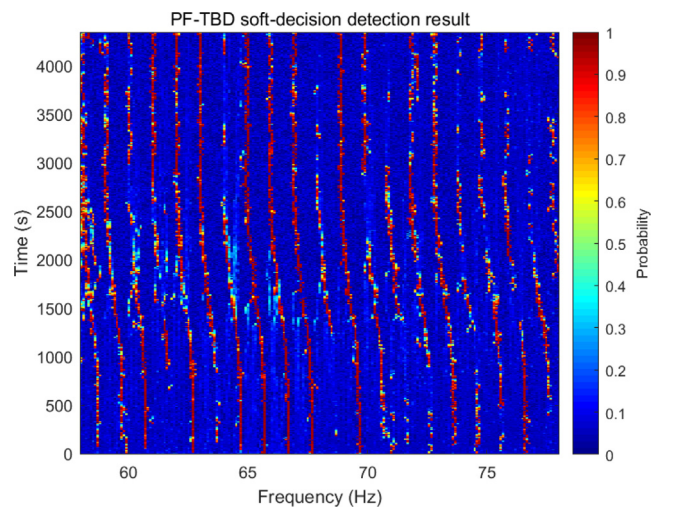


Fig. 17. PF-TBD soft-decision detection results.

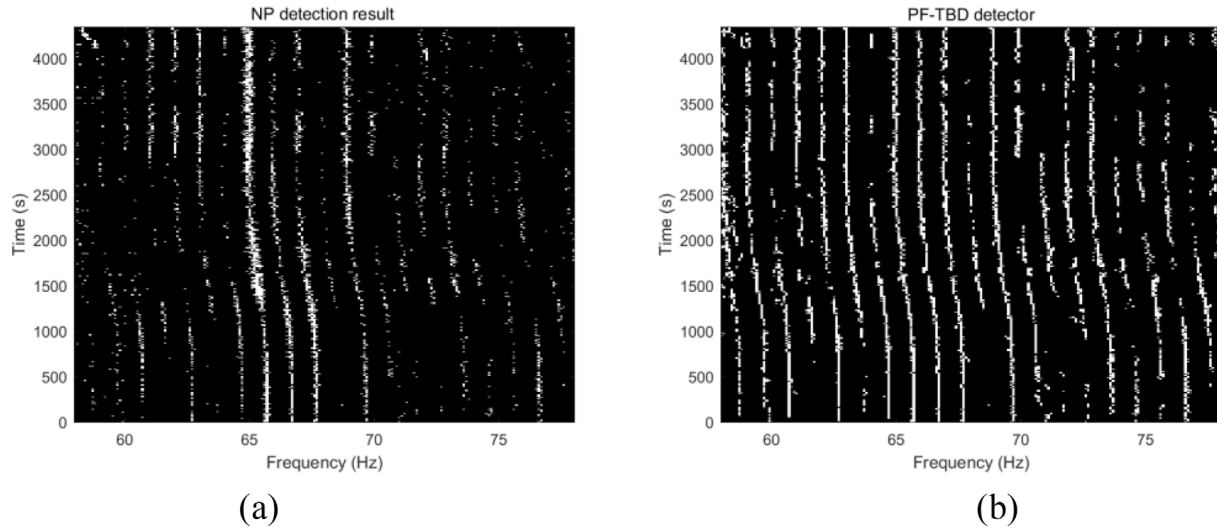


Fig. 18. (a) NP (b) PF-TBD detection results.

$$SNR = SL - TL - NL \quad (36)$$

where SL is the source level, TL is the transmission loss, and NL is the noise spectrum level. Denoting r as the range between the target and the passive sonar, there is $TL = 20\log_{10}(r)$. If $SL = 135$ dB and $NL = 75$ dB, the SNR is obtained from (36), as shown in Fig. 3 (b). Fig. 4 shows the lofargram after the 10 s DFT with sampling frequency $f_s = 15$ kHz, and the PF-TBD soft-decision detection results are shown in Fig. 5.

Fig. 5 shows that the tonal-existence probability at 100 Hz is getting higher when the target is moving towards the passive sonar. Compare the detection results between the NP detector and the PF-TBD detector, the detection results are shown in Fig. 6 when the false alarm probability is 1%. The PF-TBD shows a better detection performance when detecting a moving target's tonal. The NP detector starts to detect the tonal after 1400 s (SNR is -5 dB), but results are sporadic and unstable (the detection probability is 71.67%). The PF-TBD detector start to detect the target around 400 s (SNR is around -9 dB). Even though it is sporadic during 400–800 s, but the detection results are relatively steady after 800 s (the detection probability is 100%).

Normally, the tonal we are interested in is stable and will last for seconds. So if we can consider that the short track detection results are from the miss alarms or clutter, the results will be clearer. Let PF-TBD performs unreal-time processing (50 s and 100 s delay) to delete the short tracks of the tonal, the detection results are shown in Fig. 7.

5. Sea trial data analysis

To evaluate the proposed method, we conducted a trial to collect field data in the East China Sea. The receiver was a vector hydrophone that was set on a stationary boat, while a source was placed on a moving boat. The trial was divided into two voyages. In the first voyage, the source transmitted a continuous wave (CW) single tonal at 311 Hz. In the second voyage, the source did not transmit any signal, and the hydrophone collected the unknown tonal from the sea. Since it is hard to evaluate the SNR in the real sea trial, the SNR in this section was approximately estimated by the amplitude of tonal and noise from the data we collected.

5.1. First voyage: detecting tonals from a deterministic source

In the first voyage, the source transmitted a CW signal with a single tonal at 311 Hz and the hydrophone received the tonal and noise from the sea. The water depth was 32 m, the hydrophone depth was 17 m, and the source depth was 20 m. The hydrophone was set at the origin of the coordinate system, and the relative trajectory of the source based on GPS is shown in Fig. 8. The initial and final positions of the source were $(-2.89$ km, -0.64 km) and $(-2.89$ km, -0.24 km), respectively. Fig. 9 is the lofargram after the DFT (10 s and sampling frequency of 15 kHz). The frequency extent was 220–320 Hz, and the observation time was 1800 s. Fig. 10 shows the corresponding PF-TBD soft-decision detection results from Fig. 9. Fig. 11 shows the detection results of the NP detector and the PF-TBD detector when the false alarm probability is around 0.5%. In Fig. 11, tonal detection results at 311 Hz is 100% with both methods because the SNR at 311 Hz is high (between 8 and 15 dB). In Fig. 10 and Fig. 11b, we can see another weak tonal was detected by PF-TBD on 237 Hz from an unknown target during 1000–1400 s with SNR between -8 – 0 dB. But with NP detector, the weak tonal on 237 Hz is miss detected. Fig. 12 shows the results of the PF-TBD with unreal-time processing with delay time 200 s. The soft-decision detection results and SNR on 311 Hz and 237 Hz during the observation time are shown in Fig. 13.

5.2. Second voyage: detecting tonals from a surface vessel

In the second voyage, the hydrophone received the noise and unknown tonals from the sea. The water depth was 37 m and the hydrophone depth was 10 m. Fig. 14 shows the lofargram after the DFT (10 s and sampling frequency 15 kHz). The frequency extent was 55–75 Hz, and the observation time was 4350 s. There are several unknown observed tonals. The information from the automatic identification system (AIS) indicates that the unknown tonals were from a container vessel (Fig. 15). The relative trajectory of the container vessel from the AIS is shown in Fig. 16 when considering the hydrophone was the origin of the coordinate system. The initial and final positions of the container vessel were $(9.59$ km, 12.12 km) and $(-20.99$ km, -10.95 km), respectively.

Fig. 17 shows the corresponding proposed PF-TBD soft-decision detection results from Fig. 14. Fig. 18 illustrates that the NP detector and PF-TBD detection results when the false alarm probability is around 0.2%. To compare the detection probabilities of the two

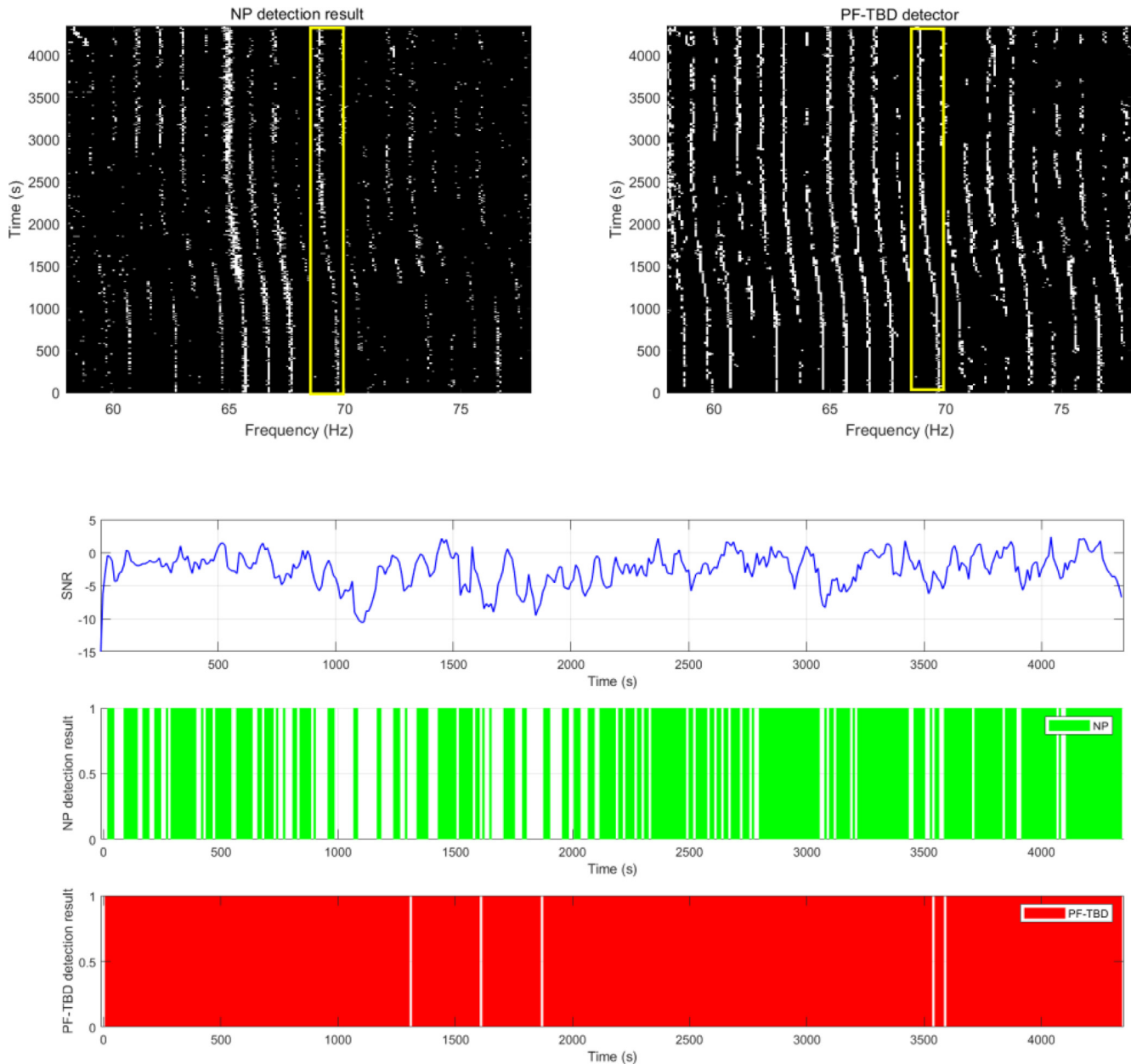


Fig. 19. NP and PF-TBD detection results and corresponding SNR for the tonal at a frequency of 68.7–69.7 Hz from 1 to 4350 s.

methods, Fig. 19 shows the tonal detection results of the NP and PF-TBD at 68.7–69.7 Hz, and the corresponding SNR. During the observation time, the SNR of the tonal is between -10 – 2 dB at 68.7–69.7 Hz. The PF-TBD shows a high probability of the tonal, the detection result is 96.55%, better than the NP detection result 69.43%.

6. Conclusion

We propose using the PF-TBD method to detect weak tonals with a soft-decision detection result. The method fully derives the statistical properties of the received signal to establish the TBD model for tonal detection. The birth and death of the tonals are presented by the soft-decision detection results at each frequency bin during the observation period. The results of simulations and two sea trials (detect tonals from a deterministic source and a surface vessel) demonstrate the feasibility of the pro-

posed method in detecting and tracking tonals. For further research, bearing estimation based on the vector hydrophone can be considered after completing tonal detection for the targets [45].

CRediT authorship contribution statement

Dianlun Zhang: Methodology, Project administration, Funding acquisition. **Linsen Gao:** Software, Formal analysis, Data curation, Writing - original draft, Methodology. **Dajun Sun:** Supervision, Conceptualization. **Tingting Teng:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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